

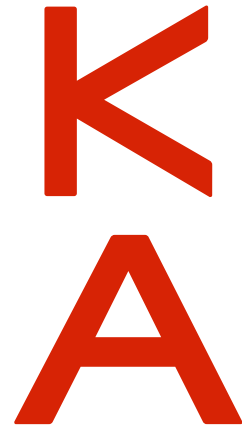


TECHNICAL WRITING

What is the definition of Technical Writing?

The Society for Technical Communication (STC) offers this definition of technical writing:

"the process of gathering information and presenting it to an audience in a clear, easily understandable form."



Characteristics of Good Technical Communication+

- **Accurate**
 - precise use of language - stylistically accurate
 - specialized technical terms - technically accurate
- **Clear**
 - simple syntax
- **Concise**
 - use of a minimum of words to express the idea
 - does not digress from the point being made
- **Coherent**
 - it develops its subject matter in an easy-to-follow line of thinking using linkers, references, etc.
- **Appropriate**
 - appropriate to its purpose and audience

Global considerations: **purpose, audience, structure**





Purpose & Corresponding Document Types

There are four general categories:

- to provide information - *reports, (literature) reviews, specifications*
- to give instructions – *instructions, procedures, work plans*
- to persuade the reader - *proposals, recommendation reports, job application letters, résumés*
- and to enact (or prohibit) something - *regulations, patents*



Main Types of Reports



Complex, expensive projects

Before: **proposal**

During: various **informational reports**

After: **completion report or recommendation report**

informational reports come in many forms



Goal: describe something that has happened or is happening now

Sometimes: **analysis** – explanation of why/how something happened, e.g. **incident report**



Main Types of Informational Reports



Type of report	Example
Directive	A supervisor explains a company's new policy
Field report	An insurance adjuster presents the results of an inspection of a building after a storm
Test report	The procedure and findings of a test are described
Progress report	Information on the project is updated
Incident report	A worker at a plant writes it after an accident
Recommendation report/Product comparison	A product comparison/analysis evaluates and contrasts two or more similar products to determine their features, specifications, performance to help the audience make informed purchasing decisions
Lab report	Present routine findings; students are asked to write them – introduction into the scientific method
Meeting minutes	a secretary produces an official record of what happened at a meeting

Technical Communication. Mike Markel, Stuart A. Selber. 2021, 13 th edition.

How to write recommendation reports?



A **recommendation report** also presents information but goes one step further by offering suggestions about what the reader should do next.

Recommendation reports might address such questions as:

- **What should we do about Problem X?**
- **Should we do Function X?**
- **Should we use Technology A or Technology B to do Function X?**
- **We currently use Method A to do Function X. Should we be using Method B?**

How to organize recommendation reports?



Recommendation reports can be a **final link in a chain** of documents that begins with a proposal and continues with one or more progress reports. This last report is often called a ***final report***, a ***project report***, a ***recommendation report***, a ***completion report***, or simply a ***report***.

A **recommendation report** can also be a **freestanding** document, e.g.: you might be asked for a recommendation on whether your company should offer employees comp time (compensating those who work overtime with time off) instead of overtime pay.

Most recommendation reports discuss the question of **feasibility**.



A Sample Recommendation Report

Company
Address



Date: December 20, 2024

To: Christine Dittrich, Head
Civil Engineering
Company

Titles are included so that readers can quickly identify the writers

From: Andrew Smith
Position, company

Subject: Progress Report for ...

! The subject heading indicates the subject and the purpose of the report

The purpose of the study...

The methods the writers used..

The principal findings: the results and the conclusions of the study...

The major recommendation. The writers ask their supervisor if she will reconsider whether the company can afford...

A polite offer to participate further or to provide more information.

(We appreciate the trust you have shown in inviting us to participate in this phase of the feasibility study, and we would look forward to working with you on any follow-up activities. If you have any questions or comments, please contact Andrew Smith, at asmith@rrvv.org or at 555-444-3322).



THE USE OF PASSIVE VOICE in REPORTS



Although writing in the *active voice*

I collected three soil samples is more

concise, clearer, and more interesting than writing in the *passive voice*

Three soil samples **were collected**,

the sciences and engineering have a long tradition of using the passive voice.

The passive voice **emphasizes the material studied and the actions taken** and **deemphasizes the role of the researcher.**

ETHICS NOTE: Presenting Data Honestly



It is unethical / dishonest

 to omit data which undercut your hypothesis

 To choose a type of graphic that obscures negative findings or

 to design a graphic so that data points are omitted



These are obvious violations of scientific norms.

Inconsistent data or contradictory results often lead researchers to examine their approach and assumptions more carefully, which can **lead to breakthroughs** in a field of study.



Identify Your Audience

- **type and level of expertise,**
- **purpose in using the document,**
- **attitude toward both you and the content of your document**

These considerations will influence specific features of the document: **organization, introduction, technical terms, and level of detail.**



Planning and Producing Documents



1. **Outline** — to organize the information
2. **Draft** — put your first draft away for a day or two if possible
3. **Edit** - Revise your document in stages:
 - organization
 - accuracy and appropriateness
 - clarity, conciseness, coherence
 - grammar, spelling, punctuation
4. **Review** - the last major step for most technical documents is one or more reviews (technical experts, technical editors, legal reviewers, supervisor).



Planning and Producing Documents



The Writing Process



Important Things to Consider

KISS principle!!

- Formal style of writing, so consequently:

No contractions! ~~It's~~ It is

No slang, colloquialisms: kind of, like...

No subjective/descriptive adjectives: great, decent

- Capitalization! Do NOT capitalize nouns.

BUT! Capitalization in titles in American English:

EVERY word should be capitalized but prepositions and

articles. E.g.: **P**roduct **R**ecall



+

K
A

Typical Mistakes in Writing



- Go easy on the Progressive!!

We very much regret that this recall ~~is happening~~.

- Articles: ~~a~~ well-known structure, ~~a~~ evolutionary project...

- Political Correctness!!

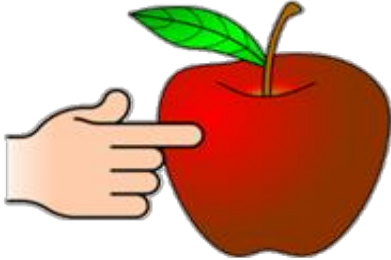
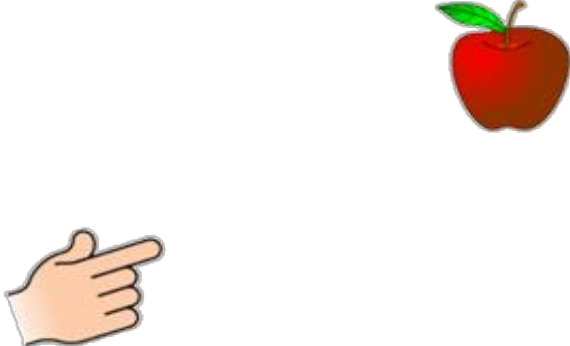
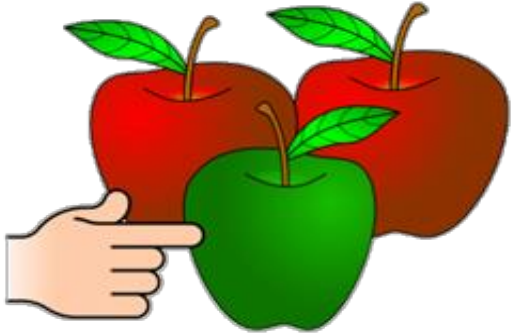
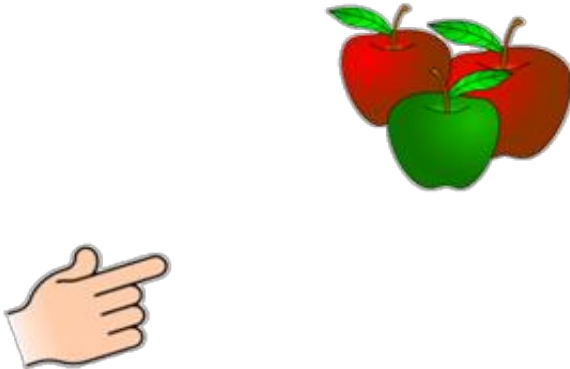
I have a question: if I write for example „everyone has their own position“, is it correct to use plural form (their) or should I use his/her/its? And what about the verb, do I say „everyone has/have“?

Customers need to contact their dealer. ~~He~~ will download the new software.



Typical Mistakes in Writing (cont.)



Singular	 <u>This</u> is an apple.	 <u>That</u> is an apple.
Plural	 <u>These</u> are apples.	 <u>Those</u> are apples.

In some circumstances, the switch between the regenerative to the ABS system may cause **this slight delays**.

Typical Mistakes in Writing (cont.)



Amount of metal, love, snow.... (uncountable)

Number of employees, managers, spare parts, cars...
(countable)

Little – less – the least (uncountable) = less metal

Few – fewer – the fewest (countable) = fewer spare parts



Paragraphing

Model X Steering Assist Motor Bolt Recall

Tesla has decided to proactively retrofit a power steering component in some Model X vehicles. This voluntary recall applies to most Model X vehicles built before mid-October 2016. Model X vehicles built after mid-October 2016 are not affected.

We are not aware of any injuries or collisions relating to the power steering component. However, we have observed excessive corrosion on the bolts that attach this component to the steering gear in affected Model X vehicles. This corrosion has been observed primarily in very cold climates that use calcium or magnesium road salts rather than sodium chloride (table salt). If the bolts fracture from corrosion, the driver may lose power steering assist. This would not prevent the driver from steering the vehicle, but it would require more force to turn the steering wheel, especially during low speed parking maneuvers when power steering assist is at its highest use. Power steering assist decreases with increasing speed, so the driver may not notice a need for more steering force at highway speeds.

If you own a Model X affected by this voluntary recall, there is no immediate action you need to take at this time, and you may continue to drive your car. Tesla will replace the bolts and apply a corrosion-preventative sealer in all affected Model X vehicles to account for the possibility that any vehicle may later be used in a cold, highly corrosive environment. The service will be free of charge. If any bolts are found broken on your vehicle or break during removal and cannot be removed, we will also replace your steering gear, again free of charge. Tesla will contact you to schedule a service appointment when parts become available in your region.

Thank you for being a Tesla customer, and we apologize for this inconvenience. If you have any questions, please [contact us](#).



Use of Topic Headings



Product Comparison

Introduction

The objective of this comparative analysis is to examine the distinctions between two types of wind turbines: the Horizontal-Axis Wind Turbine (HAWT) and the Vertical-Axis Wind Turbine (VAWT).

Similarities

Both wind turbine types incorporate an integrated gearbox and generator to harness energy generated by the rotation of the blades.

Differences in technology

The HAWT requires its rotor and shaft to be parallel to the wind, whereas the VAWT features a vertically oriented rotating shaft that is perpendicular to the wind. Furthermore, the VAWT does not necessitate a yaw system, as it can harness wind energy from any direction.

Differences in performance

There are three primary turbine types: the Savonius and Darrieus VAWTs, and the HAWT. The Savonius VAWT is the least efficient, capable of producing only half of the Betz limit. The Darrieus VAWT can capture energy from both the upwind and downwind blades, with approximately 80% of energy generated by the upwind blades and 20% by the downwind blades. Despite being somewhat more efficient than the Savonius VAWT, the Darrieus VAWT is still significantly less effective than the HAWT. Currently, vertical-axis wind turbines achieve an efficiency of around 30-40%, while horizontal-axis wind turbines can reach up to 50%.

Conclusion

Presently, HAWTs exhibit greater efficiency compared to VAWTs; however, they also come with drawbacks such as maintenance requirements, tall tower heights, and noise generation. Consequently, HAWTs are predominantly used offshore.



LANGUAGE DEVICES: Concise Technical Writing.

Hyphenated Adjectives.

Relative clauses/sentences/phrases become hyphenated adjectives.

A building developed by WS Group in Wuhan, China, is a high-rise residence with 56 stories near the Yangtze River. The building has the height of 179 m.

To investigate its seismic performance, a shaking table test with a 1/30 scale model is carried out in Structural Laboratory in Wuhan University of Technology...

(AmE story, BrE storey/floor)

A building developed by WS Group in Wuhan, China, is a high-rise 56-story 179-meter (high) residence near the Yangtze River. To investigate its seismic performance, a shaking table test with a 1/30 scale model is carried out in Structural Laboratory in Wuhan University of Technology...